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Interrupteurs à basse tension dans l'air, sectionneurs à basse tension dans l'air, interrupteurs-sectionneurs à basse tension dans l'air et combinés à fusibles à basse tension

Low-voltage air-break switches, air-break disconnectors, air-break switch-disconnectors and fuse-combination units



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**LOW-VOLTAGE AIR-BREAK SWITCHES,
AIR-BREAK DISCONNECTORS, AIR-BREAK SWITCH-DISCONNECTORS
AND FUSE-COMBINATION UNITS**

FOREWORD

- 1) The formal decisions or agreements of the IEC on technical matters, prepared by Technical Committees on which all the National committees having a special interest therein are represented, express, as nearly as possible, an international consensus of opinion on the subjects dealt with.
- 2) They have the form of recommendations for international use and they are accepted by the National Committees in that sense.
- 3) In order to promote international unification, the IEC expresses the wish that all National Committees should adopt the text of the IEC Recommendation for their national rules in so far as national conditions will permit. Any divergence between the IEC Recommendations and the corresponding national rules should, as far as possible, be clearly indicated in the latter.

PREFACE TO THE FIRST EDITION

This Recommendation has been prepared by Sub-Committee 17B, Low-Voltage Switchgear and Controlgear, of IEC Technical Committee No. 17, Switchgear and Controlgear.

Work was commenced in January 1969; two other drafts were elaborated in October 1969 and November 1970. As a result of the meeting held in Brussels in 1971, a final draft was submitted to the National Committees for approval under the Six Months' Rule in November 1971.

The following countries voted explicitly in favour of publication:

Australia	Poland
Austria	Portugal
Belgium	Romania
Czechoslovakia	South Africa
Denmark	Sweden
France	Switzerland
Germany	Turkey
Israel	Union of Soviet
Italy	Socialist Republics
Japan	United Kingdom
Netherlands	United States of America
Norway	Yugoslavia

PREFACE TO THE SECOND EDITION

This standard has been prepared by Sub-Committee 17B: Low-voltage Switchgear and Controlgear, of IEC Technical Committee No. 17: Switchgear and Controlgear.

This second edition is based on the first edition, 1972, Amendment No. 1, 1979 and Amendment No. 2, 1983.

LOW-VOLTAGE AIR-BREAK SWITCHES, AIR-BREAK DISCONNECTORS, AIR-BREAK SWITCH-DISCONNECTORS AND FUSE-COMBINATION UNITS

1. General

1.1 Scope

This standard applies to air-break switches, air-break disconnectors, air-break switch-disconnectors and fuse-combination units to be used in distribution circuits and motor circuits the rated voltage of which does not exceed 1 000 V a.c. or 1 200 V d.c.

Notes 1. — Depending on its design, a switch (or disconnector) can be referred to as “rotary switch (disconnector)”, “cam-operated switch (disconnector)”, “knife-switch (disconnector)”, etc.

2. — If they are not manually operated, switches and disconnectors may have to comply with additional requirements.

Any fuses incorporated shall conform to the requirements laid down in the relevant IEC fuse standards.

This standard does not apply to equipment coming within the scope of IEC Publication 157-1: Circuit-breakers, IEC Publication 158-1: Contactors, IEC Publication 292-1: Motor Starters, and IEC Publication 337-1: Control Switches; however, when switches and fuse-combination units coming into the scope of this standard are normally used to start, accelerate and/or stop a motor, they shall also comply with the additional requirements given in Appendix C.

This standard does not include the additional requirements necessary for electrical apparatus for explosive gas atmospheres.

Notes 3. — In this standard, the word “switch” also applies to the apparatus referred to in French as “commutateurs”, intended to modify the connections between several circuits, and *inter alia* to substitute one part of a circuit for another.

4. — As a general rule, in the rest of this standard, switches, disconnectors, switch-disconnectors and fuse-combination units will be referred to as “equipment”.

1.2 Object

The object of this standard is to state:

- 1) the characteristics of the equipment;
- 2) the conditions with which the equipment must comply with reference to:
 - a) operation and behaviour in normal service,
 - b) operation and behaviour in case of specified abnormal conditions, e.g. short circuit;
 - c) dielectric properties;
- 3) the tests intended for confirming that these conditions have been met and the methods to be adopted for these tests;
- 4) the data to be marked on the equipment.

2. Definitions

For the purpose of this standard, the following definitions shall apply:

2.1 *Devices*

2.1.1 *Switching device*

A device designed to make or break the current in one or more electric circuits.

2.1.2 *Mechanical switching device*

A switching device designed to close and open one or more electric circuits by means of separable contacts.

2.1.3 *Switch (mechanical)*

A mechanical switching device capable of making, carrying and breaking currents under normal circuit conditions which may include specified operating overload conditions and also carrying for a specified time currents under specified abnormal circuit conditions such as those of short circuit.

Note. — A switch may also be capable of making, but not breaking, short-circuit currents.

2.1.4 *Disconnecter (isolator)*

A mechanical switching device which, for reasons of safety, provides in the open position an isolating distance in accordance with specified requirements.

A disconnector is capable of opening and closing a circuit when either negligible current is broken or made, or when no significant change in the voltage across the terminals of each of the poles of the disconnector occurs. It is also capable of carrying currents under normal circuit conditions and carrying for a specified time currents under abnormal conditions such as those of short circuit.

Notes 1. — "Negligible currents" imply currents such as the capacitance currents of bushings, busbars, connections, very short lengths of cables and currents of voltage transformers and dividers.

"No significant change in voltage" refers to such applications as the by-passing of induction voltage regulators or circuit-breakers.

2. — There are devices, referred to in this standard as disconnectors, which possess a certain making and/or breaking capacity.

2.1.5 *Switch-disconnector (switch-isolator)*

A switch which, in the open position, satisfies the isolating requirements specified for a disconnector.

2.1.6 *Fuse*

A switching device that, by the melting of one or more of its specially designed and proportioned components, opens the circuit in which it is inserted and breaks the current when this exceeds a given value for a sufficient time.

Note. — The fuse comprises all the parts that form the complete switching device.

2.1.7 *Fuse-combination unit*

A combination of a switch, a disconnector or a switch-disconnector and one or more fuses in a composite unit, made by the manufacturer or in accordance with his instructions.

2.1.8 *Switch-fuse*

A switch in which one or more poles have a fuse in series in a composite unit.

2.1.9 *Disconnecter-fuse*

A disconnector in which one or more poles have a fuse in series in a composite unit.

2.1.10 *Fuse-switch*

A switch in which a fuse-link or a fuse carrier with fuse-link forms the moving contact of the switch.

2.1.11 *Fuse-disconnector (fuse-isolator)*

A disconnector in which a fuse-link or a fuse carrier with fuse-link forms the moving contact of the disconnector.

2.2 *Constructional elements*

2.2.1 *Conductive part*

A part which is capable of conducting current although it may not necessarily be used for carrying service current.

2.2.2 *Exposed conductive part*

A conductive part which can be touched readily and which normally is not live, but which may become live under fault conditions.

Note. — Typical exposed conductive parts are walls of enclosures, operating handles, etc.

2.2.3 *Live part*

Any conductor or conductive part which is under a potential in normal use.

Note. — Live parts also include the neutral conductor and conductive parts which are connected with it.

2.2.4 *Main circuit (of a switching device)*

All the conductive parts of a switching device included in the circuit which it is designed to close or open.

2.2.5 *Pole of a switching device*

The portion of a switching device associated exclusively with one electrically separated conducting path of its main circuit and excluding those portions which provide a means for mounting and operating all poles together.

Note. — A switching device is called single-pole if it has only one pole. If it has more than one pole, it may be called multipole (two-pole, three-pole, etc.) provided the poles are or can be coupled in such a manner as to operate together.

2.2.6 *Main contact (of a mechanical switching device)*

A contact included in the main circuit of a mechanical switching device, intended to carry in the closed position the current of the main circuit.

2.2.7 *Fuse-link (fuse-unit)*

The part of a fuse intended to be replaced after the fuse has operated.

2.3 *Conditions of operation*

2.3.1 *Operation (of a mechanical switching device)*

The transfer of the moving contact(s) from one position to an adjacent position.

Notes 1. — This may be a closing operation or an opening operation.

2. — If distinction is necessary, an operation in the electrical sense, e.g. make or break, is referred to as a switching operation and an operation in the mechanical sense, e.g. close or open, is referred to as a mechanical operation.

2.3.2 *Operating cycle (of a mechanical switching device)*

A succession of operations from one position to another and back to the first position through all other positions, if any.

2.3.3 *Dependent manual operation (of a mechanical switching device)*

An operation solely by means of directly applied manual energy, such that the speed and force of the operation are dependent upon the action of the operator.

2.3.4 *Stored energy operation (of a mechanical switching device)*

An operation by means of energy stored in the mechanism itself prior to the completion of the operation and sufficient to complete it under pre-determined conditions.

Note. — This kind of operation may be subdivided according to:

- 1) how the energy is stored (spring, weight, etc.);
- 2) how the energy originates (manual, electric, etc.);
- 3) how the energy is released (manual, electric, etc.).

2.3.5 *Independent manual operation (of a mechanical switching device)*

A stored energy operation where the energy originates from manual power, stored and released in one continuous operation, such that the speed and force of the operation are independent of the action of the operator.

2.4 *Characteristic quantities*

2.4.1 *Rated value*

A stated value of any one of the characteristic quantities that serve to define the working conditions for which the equipment is designed and built.

2.4.2 *Utilization category (for a switching device)*

A combination of specified requirements, related to operating conditions, and selected to represent a significant group of practical applications.

Note. — The requirements to be specified may include such items as values of making and breaking capacities, the associated circuits, and the relevant conditions of use and behaviour.

2.4.3 *Prospective current (of a circuit, and with respect to an equipment)*

The current that would flow in the circuit, if each pole of the equipment were replaced by a conductor of negligible impedance.

Note. — The prospective current may be qualified in the same manner as an actual current, e.g.: prospective breaking current, prospective peak current, etc.

2.4.4 *Making capacity*

A value of current that a switch or a switch-disconnector is capable of making at a stated voltage and under prescribed conditions of use and behaviour.

For a.c., it is the r.m.s. value of the a.c. component.

2.4.5 *Short-circuit making capacity*

A value of prospective peak current that a switch or a switch-disconnector is capable of making at a stated voltage and under prescribed conditions including a short circuit at the terminals of the apparatus.

2.4.6 *Breaking capacity*

A value of breaking current that a switch or a switch-disconnector is capable of breaking at a stated voltage and under prescribed conditions of use and behaviour.

For a.c., it is the r.m.s. value of the a.c. component.

2.4.7 *Short-time withstand current*

The current that a switch, a disconnector or a switch-disconnector can carry in the closed position during a specified short time under prescribed conditions of use and behaviour.

2.4.8 *Applied voltage*

The voltage which exists across the terminals of a pole of a switching device just before the making of the current.

2.4.9 *Recovery voltage*

The voltage which appears across the terminals of a pole of a switching device after the breaking of the current.

Note. — This voltage may be considered in two successive intervals of time, one during which a transient voltage exists, followed by a second one during which power-frequency voltage (Sub-clause 2.4.9.2) alone exists.

2.4.9.1 *Transient recovery voltage*

The recovery voltage during the time in which it has a significant transient character.

Notes 1. — The transient voltage may be oscillatory or non-oscillatory or a combination of these depending on the characteristics of the circuit and the switching device. It includes the voltage shift of the neutral of a polyphase circuit.

2. — The transient recovery voltage in three-phase circuits is, unless otherwise stated, that across the first pole to clear because this voltage is generally higher than that which appears across each of the other two poles.

2.4.9.2 *Power-frequency recovery voltage*

The recovery voltage after the transient voltage phenomena have subsided.

Note. — This definition applies also to the case of d.c., the frequency then being considered as zero.

2.4.10 *Clearance*

The distance between two conductive parts along a string stretched the shortest way between these conductive parts.

2.4.10.1 *Clearance between poles*

The clearance between any conductive parts of adjacent poles.

2.4.10.2 *Clearance to earth*

The clearance between any conductive parts and any parts which are earthed or intended to be earthed.

2.4.10.3 *Clearance between open contacts (gap)*

The total clearance between the contacts, or any conductive parts connected thereto, of a pole of a mechanical switching device in the open position.

2.4.11 *Creepage distance*

The shortest distance along the surface of an insulating material between two conductive parts.

Note. — A joint between two pieces of insulating material is considered part of the surface.

2.4.12 *Isolating distance (of a pole of a mechanical switching device)*

The clearance between open contacts meeting the safety requirements specified for disconnectors.

3. **Classification**

3.1 According to the method of operation of manually operated equipment, these are designated as having:

- dependent manual operation (see Sub-clause 2.3.3);
- independent manual operation (see Sub-clause 2.3.5).

3.2 According to the degree of protection provided by the enclosure, distinction is made in accordance with IEC Publication 144: Degrees of Protection of Enclosures for Low-voltage Switchgear and Controlgear.

4. Characteristics of equipment

4.1 Summary of characteristics

The characteristics of the equipment shall be stated in the following terms, where such terms are applicable:

Type of equipment (see Sub-clause 4.2);

Rated values (see Sub-clause 4.3);

Auxiliary circuits (see Sub-clause 4.4);

Degrees of protection provided by enclosures (see IEC Publication 144).

4.2 Type of equipment

The following shall be stated:

4.2.1 Number of poles

4.2.2 Number of positions (if more than two)

4.2.3 Kind of current

Kind of current (a.c. or d.c.) and, in the case of a.c., number of phases and rated frequency.

4.3 Rated values

Rated values are assigned by the manufacturer. They shall be stated in accordance with Sub-clauses 4.3.1 to 4.3.13, but it is not necessary to establish all the rated values listed.

Note. — Switches, switch-disconnectors and fuse-combination units normally intended for direct switching of individual motors shall also comply with the additional requirements of Appendix C.

4.3.1 Rated voltages

An equipment is defined by the following rated voltages:

4.3.1.1 Rated operational voltages

A rated operational voltage (U_o) of an equipment is a value of voltage which, combined with a rated operational current, determines the application of the equipment and to which are referred the making and breaking capacities, and the utilization category.

For polyphase circuits, it is stated as the voltage between phases.

Note. — An equipment may be assigned a number of combinations of rated operational voltages and rated operational currents for different utilization categories.

4.3.1.2 Rated insulation voltage

The rated insulation voltage (U_i) of an equipment is the value of voltage which designates it and to which dielectric tests, clearances and creepage distances are referred.

Unless otherwise stated, the rated insulation voltage is the value of the maximum rated operational voltage of the equipment. In no case, shall the maximum rated operational voltage exceed the rated insulation voltage.

4.3.2 Rated currents

An equipment is defined by the following rated currents:

4.3.2.1 *Rated conventional thermal current*

The rated conventional thermal current (I_{th}) of an equipment is the maximum current stated by the manufacturer that the unenclosed equipment can carry in 8 h duty (see Sub-clause 4.3.4.1) when tested in free air, without the temperature-rise of its several parts exceeding the limits specified in Sub-clause 7.3 (Table V) when tested according to Sub-clause 8.2.2.

Notes 1. — Free air is understood to be that obtained under normal indoor conditions reasonably free from dust and external radiation.

2. — Unenclosed equipment is equipment supplied by the manufacturer without an enclosure or equipment supplied by the manufacturer with an enclosure forming an integral part of the equipment.

4.3.2.2 *Rated enclosed thermal current*

The rated enclosed thermal current (I_{the}) of an equipment is the maximum current stated by the manufacturer that the equipment can carry in the stated duty (see Sub-clause 4.3.4) when mounted in a specified enclosure. Tests for this rating shall be in accordance with Sub-clause 8.2.2, but are not mandatory if the test for "rated conventional thermal current" has been made, and the manufacturer is prepared to state an enclosed thermal current rating.

The rating may be an unventilated rating, in which case the enclosure shall be of the size stated by the manufacturer to be the smallest enclosure that is applicable in service. Alternatively, the rating may be a ventilated rating with the ventilation in accordance with the manufacturer's data.

Note. — It is not possible to usefully define a service thermal current rating as the installation and service conditions can vary greatly. (The "rated current" Sub-clause 4.2 of IEC Publication 439 is in effect the "service thermal current rating".)

For fuse-combination units, the temperature-rises of the fuse-link contacts shall not exceed the value which ensures compliance with Notes 1 and 2 of Table V (see Sub-clause 7.3) with respect to the fuse elements; the manufacturer of the fuse-combination unit shall state the type of fuse-links to be used.

4.3.2.3 *Rated operational currents or rated operational powers*

A rated operational current (I_e) of an equipment is stated by the manufacturer and takes into account the rated operational voltage (see Sub-clause 4.3.1.1), the rated frequency (see Sub-clause 4.3.3), the rated duty (see Sub-clause 4.3.4), the utilization category (see Sub-clause 4.3.11) and the type of protective enclosure.

In the case of equipment for direct switching of individual motors, the indication of a rated operational current may be replaced or supplemented by the indication of the maximum rated power output, at the rated operational voltage considered, of the motor for which the equipment is intended. The manufacturer shall be prepared to state the relationship assumed between the current and the power.

4.3.3 *Rated frequency*

The supply frequency for which an equipment is designed and to which the other characteristic values correspond.

4.3.4 *Rated duty*

The rated duties considered as normal are as follows:

4.3.4.1 *Eight-hour duty*

Duty in which the main contacts of an equipment remain closed, whilst carrying a steady current long enough for the equipment to reach thermal equilibrium but not for more than eight hours without interruption.

Notes 1. — This is the basic duty on which the rated thermal current of the equipment is determined.

2. — Interruption means breaking of the current by operation of the equipment.

4.3.4.2 *Uninterrupted duty*

Duty in which the main contacts of an equipment remain closed, whilst carrying a steady current without interruption for periods of more than eight hours (weeks, months, or even years).

Note. — This kind of service is set apart from the eight-hour duty because oxides and dirt can accumulate on the contacts and lead to progressive heating. Uninterrupted duty can be taken account of either by a derating factor, or by special design consideration (e.g.: silver contacts) (see Table V).

4.3.5 *Rated making capacity*

The rated making capacity of a switch or a switch-disconnector is a value of current, stated by the manufacturer, determined under steady-state conditions, which the apparatus can satisfactorily make under specified making conditions.

The making conditions which shall be specified are:

- the voltage between poles before contact making;
- the characteristics of the test circuit.

The rated making capacity is stated by reference to the rated operational voltage and rated operational current and to the utilization category, according to Table II.

It is recalled that, for a.c., the rated making capacity is expressed by the r.m.s. value of the a.c. component.

Note. — In the case of disconnectors having a making capacity although they be of utilization category AC-20 or DC-20, this value shall be stated separately by the manufacturer.

The rated making capacity is only valid when the apparatus is operated in accordance with the requirements of Sub-clause 7.5.

4.3.6 *Rated breaking capacity*

The rated breaking capacity of a switch or a switch-disconnector is a value of current, stated by the manufacturer, which the apparatus can satisfactorily break under specified breaking conditions at the rated operational voltage.

The breaking conditions which shall be specified are:

- the characteristics of the test circuit;
- the recovery voltage.

The rated breaking capacity is stated by reference to the rated operational voltage and rated operational current and to the utilization category, according to Table II.

An apparatus shall be capable of breaking any value of the load current up to its highest rated breaking capacity according to Sub-clause 4.3.11.

It is recalled that, for a.c., the rated breaking capacity is expressed by the r.m.s. value of the a.c. component.

Note. — In the case of disconnectors having a breaking capacity although they be of utilization category AC-20 or DC-20, this value shall be stated separately by the manufacturer.

The rated breaking capacity is only valid when the apparatus is operated in accordance with the requirements of Sub-clause 7.5.

4.3.7 *Rated short-time withstand current*

The rated short-time withstand current of a switch, a disconnector or a switch-disconnector is the value of short-time withstand current that the apparatus can carry without damage (see Sub-clause 8.2.5).

Unless otherwise stated by the manufacturer, the duration is one second, and the value of the current is twenty times the maximum rated operational current.

Note. — For values different from these, the manufacturer shall state both the value of the short-time withstand current and the duration.

For a.c., the value of the current is the r.m.s. value of the a.c. component and it is assumed that the highest peak value likely to occur does not exceed n times this r.m.s. value, the factor n being given by Table X.

4.3.8 *Rated short-circuit making capacity*

The rated short-circuit making capacity of a switch or a switch-disconnector is the value of short-circuit making capacity assigned to the apparatus by the manufacturer for the rated operational voltage, at rated frequency (if any) and at a specified power-factor (or time-constant). It is expressed as the maximum prospective peak current.

For a.c., the relationship between power-factor, prospective peak current and r.m.s. current shall be in accordance with Table X.

4.3.9 *Rated conditional short-circuit current*

The rated conditional short-circuit current of an equipment is the value of prospective current, stated by the manufacturer, which the equipment, protected by a specified current-limiting switching device, can satisfactorily withstand for the operating time of this device under the test conditions specified in Sub-clause 8.2.6.

Notes 1. — For a.c., the rated conditional short-circuit current is expressed by the r.m.s. value of the a.c. component.

2. — The current-limiting switching device may either form an integral part of the equipment or be a separate unit.

4.3.10 *Rated fused short-circuit current*

The rated fused short-circuit current of an equipment is the rated conditional short-circuit current when the current-limiting switching device is a fuse (see Sub-clause 8.2.7).

The details of the fuse shall be stated by the manufacturer.

4.3.11 *Utilization category*

The utilization categories as given in Table I and Table C-I (see Appendix C) are considered standard in this standard.

Note. — The utilization categories referred to in Tables I and II do not apply to an equipment normally used to start, accelerate and/or stop individual motors. The utilization categories for such an equipment are dealt with in Appendix C (see Sub-clause C-4.3.6).

Each utilization category is characterized by the values of the currents and voltages, expressed as multiples of the rated operational current and the rated operational voltage, as well as the power-factors or time-constants of the circuit. The conditions for making and breaking given in Table II correspond in principle to the applications listed in Table I.

TABLE I
Utilization categories

Nature of current	Category	Typical applications
Alternating current	AC-20	Connecting and disconnecting under no-load conditions.
	AC-21	Switching of resistive loads, including moderate overloads.
	AC-22	Switching of mixed resistive and inductive loads, including moderate overloads.
	AC-23	Switching of motor loads or other highly inductive loads.
Direct current	DC-20	Connecting and disconnecting under no-load conditions.
	DC-21	Switching of resistive loads, including moderate overloads.
	DC-22	Switching of mixed resistive and inductive loads, including moderate overloads (e.g.: shunt motors).
	DC-23	Switching of highly inductive loads (e.g.: series motors).

Note. — The switching of capacitors or of tungsten filament lamps shall be subject to special agreement between manufacturer and user.

TABLE II

Verification of rated making and breaking capacities (see Sub-clause 8.2.4)
Conditions for making and breaking corresponding to the several utilization categories¹⁾

	Category	Value of the rated operational current	Make			Break		
			I/I_e	U/U_e	$\cos \varphi$	I_c/I_e	U_r/U_e	$\cos \varphi$
A.C.	AC-20 AC-21 AC-22 AC-23	(All values) (All values) (All values) $I_e \leq 17 \text{ A}$ $17 \text{ A} < I_e \leq 100 \text{ A}$ $I_e > 100 \text{ A}$	I/I_e	U/U_e	$\cos \varphi$	I_c/I_e	U_r/U_e	$\cos \varphi$
			2)	1.1	2)	2)	1.1	2)
			1.5	1.1	0.95	1.5	1.1	0.95
			3	1.1	0.65	3	1.1	0.65
			10	1.1	0.65	8	1.1	0.65
			10	1.1	0.35	8	1.1	0.35
D.C.	DC-20 DC-21 DC-22 DC-23	(All values) (All values) (All values) (All values)	I/I_e	U/U_e	L/R (ms)	I_c/I_e	U_r/U_e	L/R (ms)
			2)	1.1	2)	2)	1.1	2)
			1.5	1.1	1	1.5	1.1	1
			4	1.1	2.5	4	1.1	2.5
			4	1.1	15	4	1.1	15

I_e Rated operational current (see Sub-clause 4.3.2.3);
 U_e Rated operational voltage (see Sub-clause 4.3.1.1);
 I Current made;

U Voltage before make;
 U_r Recovery voltage;
 I_c Current broken.

¹⁾ For a.c., the conditions for making are expressed in r.m.s. values, but it is understood that the peak value of asymmetrical current, corresponding to the power-factor of the circuit, may assume a higher value.

²⁾ If the making and/or breaking capacities are different from zero, the values of current and the power-factor (time-constant) shall be stated by the manufacturer.

³⁾ With a minimum of 1000 A for I .

⁴⁾ With a minimum of 800 A for I_c .

4.3.12 Mechanical endurance

With respect to its resistance to mechanical wear, an equipment is characterized by the number, stated by the manufacturer, of no-load operating cycles (i.e., without current on the main contacts) which can be made before it becomes necessary to service or replace any mechanical parts; however, normal maintenance according to the manufacturer's instructions is permitted.

The preferred numbers of no-load operating cycles are:
300, 1 000, 3 000, 10 000, 30 000, 100 000, 300 000 and 1 000 000.

Each operating cycle consists of one closing operation followed by one opening operation.

If the number of operating cycles is not stated by the manufacturer, the values in Table III shall apply.

TABLE III

Number of mechanical operating cycles, unless otherwise stated

Rated operational current I_e (A)	Number of mechanical operating cycles
$0 < I_e \leq 63$	10 000
$63 < I_e \leq 250$	3 000
$250 < I_e \leq 800$	1 000
$800 < I_e$	300

4.3.13 Electrical endurance

With respect to its resistance to electrical wear, an equipment is characterized by the number of on-load operating cycles, corresponding to the service conditions given in Table IV, which can be made without repair or replacement.

Unless otherwise stated by the manufacturer, the number of on-load operating cycles for all utilization categories, except AC-20 and DC-20, shall be not less than 1/20 of the number of no-load operating cycles corresponding to the mechanical endurance of the equipment. For the case of utilization categories AC-20 and DC-20, when Note ²⁾ of Table II (see Sub-clause 4.3.11) applies, the number of on-load operating cycles shall be stated by the manufacturer.

TABLE IV

*Verification of the number of on-load operating cycles
Conditions for making and breaking corresponding to the several utilization categories¹⁾*

	Category	Value of the rated operational current	Make			Break		
			I/I_e	U/U_e	$\cos \varphi$	I_c/I_e	U_r/U_e	$\cos \varphi$
A.C.	AC-21	(All values)	1	1	0.95	1	1	0.95
	AC-22	(All values)	1	1	0.65	1	1	0.65
	AC-23 {	$I_e \leq 17 \text{ A}$	1	1	0.65	1	1	0.65
		$I_e > 17 \text{ A}$	1	1	0.35	1	1	0.35
D.C.	DC-21	(All values)	I/I_e	U/U_e	L/R (ms)	I_c/I_e	U_r/U_e	L/R (ms)
	DC-22	(All values)	1	1	1	1	1	1
	DC-23	(All values)	1	1	2	1	1	2
		(All values)	1	1	7.5	1	1	7.5

I_e Rated operational current (see Sub-clause 4.3.2.3);
 U_e Rated operational voltage (see Sub-clause 4.3.1.1);
 I Current made;
 U Voltage before make;
 U_r Recovery voltage;
 I_c Current broken.

¹⁾ For a.c., the conditions for making are expressed in r.m.s. values, but it is understood that the peak value of asymmetrical current, corresponding to the power-factor of the circuit, may assume a higher value.

4.4 Auxiliary circuits

The characteristics of auxiliary circuits are:

- the number of those circuits;
- the number and kind of contacts (a-contact, b-contact, etc.);
- for each of these circuits:
 - rated voltage;
 - rated frequency, if any;
 - rated current;
 - rated making and breaking capacities of the contacts.

Unless otherwise stated, the rated current of auxiliary circuits is 6 A and the rated voltage and frequency (if any) of the auxiliary circuits are the rated voltage and frequency of the main circuit.

5. Markings

Each equipment shall be marked in a durable manner with the following data. The markings shall be on the equipment itself or on a nameplate or nameplates attached to the equipment, and shall be located in a place such that they are visible and legible when the equipment is installed:

- the manufacturer's name or trade mark;
- type designation or serial number;
- number of this standard;
- rated operational voltages (see Sub-clause 4.3.1.1);

- e) utilization category and rated operational currents (or rated powers), at the rated operational voltages of the equipment (see Sub-clause 4.3.2.3);
- f) either value of the rated frequency, e.g. ~ 50 Hz, or the indication "d.c." (or the symbol $\equiv$).

If not evident from information stated elsewhere by the manufacturer, the following shall also be stated on the equipment nameplate:

- g) rated insulation voltage (see Sub-clause 4.3.1.2);
- h) rated thermal current (see Sub-clauses 4.3.2.1 and 4.3.2.2);
- i) rated duty (see Sub-clause 4.3.4);
- j) rated short-time withstand current together with its duration, if applicable (see Sub-clause 4.3.7);
- k) rated short-circuit making capacity, if applicable (see Sub-clause 4.3.8);
- l) rated conditional short-circuit current, if applicable (see Sub-clause 4.3.9);
- m) rated fused short-circuit current with prescribed fuse-link(s), if applicable (see Sub-clause 4.3.10);
- n) IP number, in case of an enclosed equipment.

Note. — If the available space is insufficient to carry all the above data, the equipment shall carry at least the information under a) and b) permitting the complete data (in particular, in case of a fuse-combination unit, the relevant characteristics of the fuse-links to be used) to be obtained from the manufacturer.

6. Standard conditions for operation in service

6.1 Normal service conditions

Equipment complying with this standard shall be capable of operating under the following standard conditions.

For non-standard conditions in service, see Appendix A.

6.1.1 Ambient air temperature

The ambient air temperature does not exceed $+40^{\circ}\text{C}$ and its average over a period of 24 h does not exceed $+35^{\circ}\text{C}$.

The lower limit of the ambient air temperature is -5°C .

Note. — Equipment intended to be used in ambient air temperatures above $+40^{\circ}\text{C}$ (e.g. in forges, boiler rooms, tropical countries) or below -5°C shall be designed or used according to an agreement between manufacturer and user. Information given in the manufacturer's catalogue may take the place of such an agreement.

6.1.2 Altitude

The altitude of the site of installation does not exceed 2 000 m (6 600 ft).

Note. — For installations at higher altitudes, it is necessary to take into account the reduction of the dielectric strength and of the cooling effect of the air. Equipment so used shall be designed or used according to an agreement between manufacturer and user. Information given in the manufacturer's catalogue may take the place of such an agreement.

6.1.3 Atmospheric conditions

The air is clean and its relative humidity does not exceed 50% at a maximum temperature of $+40^{\circ}\text{C}$. Higher relative humidities may be permitted at lower temperatures, e.g. 90% at $+20^{\circ}\text{C}$. Care should be taken of moderate condensation which may occasionally occur due to variations in temperature.

6.1.4 *Conditions of installation*

The equipment is installed in accordance with the manufacturer's instructions.

7. **Standard conditions for construction**

7.1 *Mechanical design*

7.1.1 *General*

Materials shall be suitable for the particular application.

The closed and the open positions of the equipment shall be definite and shall be clearly indicated either adjacent to the actuator or by the position of the latter. If symbols are used, "I" and "O" shall be used to indicate the closed and open positions respectively.

7.1.2 *Actuator (operating handle)*

The actuator of the equipment shall be insulated from the live parts for the rated insulation voltage.

Moreover:

- if it is made of metal, it shall be capable of being satisfactorily connected to a protective conductor unless it is provided with additional reliable insulation;
- if it is made of or completely covered by insulating material, any internal metal parts, which might become accessible in the event of insulation damage, shall also be insulated from live parts for the rated insulation voltage or shall be capable of being satisfactorily connected to a protective conductor.

7.1.3 *Clearances and creepage distances*

See Appendix B.

7.1.4 *Terminals*

Terminal connections shall be such that the conductors may be connected by means of screws or other equivalent means so as to ensure that the necessary contact pressure is maintained permanently.

Terminals shall be so designed that they clamp the conductor between metal surfaces with sufficient contact pressure and without significant damage to the conductor.

Terminals shall not allow the conductors to be displaced, or be displaced themselves in a manner detrimental to the operation or the insulation (by reducing clearances and/or creepage distances).

7.1.4.1 *Arrangement of terminals*

The terminals intended for the connection of external conductors shall be so arranged that they are readily accessible under the intended conditions of use.

7.1.4.2 *Provisions for earthing*

a) *Constructional requirements*

The exposed conductive parts (for example chassis, framework and fixed parts of metal enclosures) other than those which do not constitute a danger shall be electrically interconnected and connected to a protective earth terminal for connection to an earth electrode or to an external protective conductor.

This requirement can be met by the normal structural parts providing adequate electrical continuity and applies whether the equipment is used on its own or incorporated in an assembly.

Exposed conductive parts are considered not to constitute a danger if they cannot be touched on large surfaces or grasped with the hand or if they are of small size (approximately 50 mm × 50 mm) or are so located as to exclude any contact with live parts.

Examples of these are screws, rivets, nameplates, electromagnets of relays and certain parts of releases, irrespective of their size.

b) Protective earth terminal

The protective earth terminal shall be readily accessible and so placed that the connection of the equipment to the earth electrode or to the protective conductor is maintained when the cover or any removable part is removed.

Under no circumstance shall a removable metal part of the enclosure be insulated from the part carrying the earth terminal when the removable part is in place.

The protective earth terminal shall be suitably protected against corrosion.

c) Marking and identification

This terminal shall be clearly and permanently identified by its shape, its location or its marking.

As far as marking is concerned, the identification shall be achieved by colour (green-yellow mark) or by the notation PE according to Sub-clause 5.3 of IEC Publication 445: Identification of Apparatus Terminals and General Rules for a Uniform System of Terminal Marking, Using an Alphanumeric Notation, or by a graphical symbol for use on equipment.

The graphical symbol to be used is the symbol

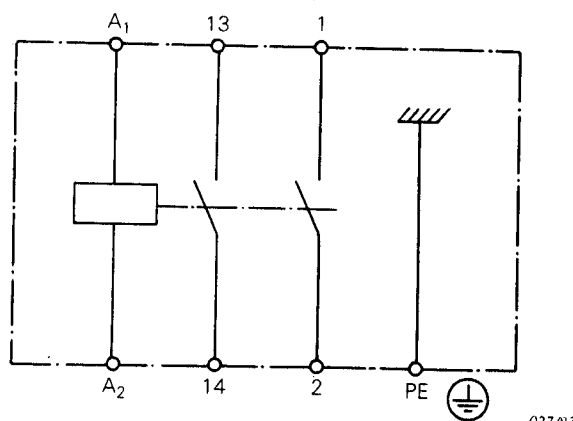
417-IEC-5019-a  Protective earth (ground)

in compliance with IEC Publication 417: Graphical Symbols for Use on Equipment. Index, Survey and Compilation of the Single Sheets.

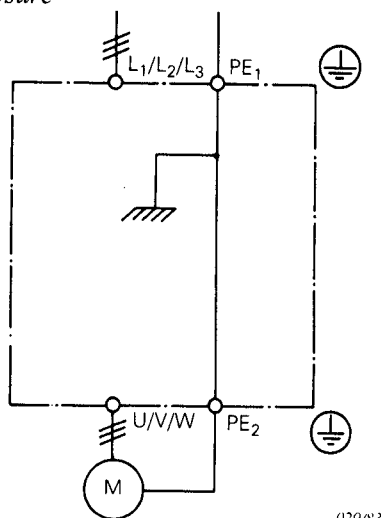
Note. — The symbol  (417-IEC-5017-a), previously recommended, shall be progressively superseded by the preferred symbol 417-IEC-5019-a, given above.

d) Examples of marking

1. Equipment

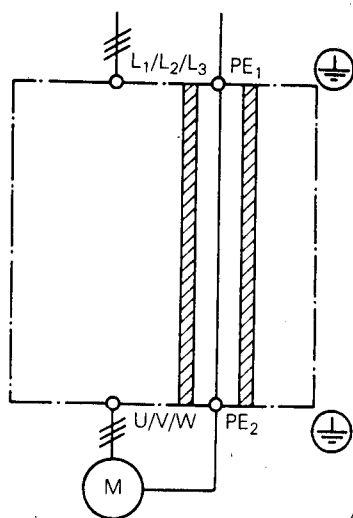


2. Equipment in metallic enclosure



029/83

3. Equipment in insulated enclosure



028/83

7.2 Enclosures

If there is an enclosure, it shall be so arranged that when it is opened, the terminals as well as all parts requiring maintenance, as prescribed by the manufacturer, are readily accessible.

Sufficient space shall be left in the interior of the enclosure for the accommodation of external conductors from their point of entry into the enclosure as far as the terminals.

The movable parts of the protective enclosure shall be firmly secured to the fixed parts by a device such that they cannot be accidentally loosened or detached owing to the effects of operation of the equipment.

When an enclosure is so designed as to allow the covers to be opened without the use of tools, means shall be provided to prevent loss of the fastening devices.

7.3 Temperature rise

7.3.1 Results to be obtained

The temperature rise of the several parts of an equipment, measured during a test carried out under the conditions specified in Sub-clause 8.2.2, shall not exceed the limiting values stated in Table V.

7.3.2 Ambient air temperature

The temperature-rise limits given in Table V are applicable only if the ambient air temperature remains within the limits given in Sub-clause 6.1.1.

7.4 Dielectric properties

The equipment shall be capable of withstanding the dielectric tests specified in Sub-clause 8.2.3.

7.5 Operating conditions

The equipment shall be operated in accordance with the manufacturer's instructions, especially for equipment with dependent manual operation where the making and breaking capacities may depend on the skill of the operator.

7.6 Additional safety requirements for disconnectors and switch-disconnectors

Disconnectors and switch-disconnectors shall provide in the open position an isolating distance in accordance with specified requirements (see Appendix B) and shall be fitted with a reliable indicating device, indicating the positions of the moving contacts. This position indicator shall be connected to the moving contacts in a reliable way; the handle may form such an indicator, provided it cannot indicate the OFF position unless all the moving contacts are in the open position.

Such an indicator is not required if the contact separation is externally visible.

TABLE V
Temperature-rise limits for the various materials and parts

Type of material Description of part	Temperature-rise limit
Contact parts (main and auxiliary contacts):	
– copper { uninterrupted duty	45 °C
{ eight-hour duty	65 °C
– silver or silver-faced*	1)
– all other metals or sintered metals	2)
Bare conductors	1)
Metallic parts acting as springs	3)
Metallic parts in contact with insulating materials	4)
Terminals for external insulated connections	70 °C ⁵⁾
Manual operating means (handles, levers, wheels, etc.):	
– parts of metal	15 °C
– parts of insulating material	25 °C
* The expression "silver-faced" includes solid silver inserts as well as electrolytically deposited silver, provided that, in the latter case, the temperature-rise requirements of Sub-clause 8.2.10.6 are satisfied. Contacts faced with other materials, the contact resistance of which is not significantly altered by oxidation, are treated as silver-faced contacts. 1) Limited solely by the necessity of not causing any damage to adjacent parts. 2) To be specified according to the properties of the metals used and limited by the necessity of not causing any damage to adjacent parts. 3) The resulting temperature should not reach a value such that the elasticity of the material is impaired. For pure copper, this implies a total temperature not exceeding +75 °C. 4) Limited solely by the necessity of not causing any damage to insulating materials. 5) The temperature-rise limit of 70 °C is a value based on the conventional test of Sub-clause 8.2.2.2. An equipment used or tested under installation conditions may have connections the type, nature and disposition of which will not be the same as those adopted for the test; a different temperature rise of terminals may result and it may be required or accepted.	

8. Tests

8.1 *Verification of the characteristics of equipment*

The tests to verify the characteristics of equipment comprise:

- type tests (see Sub-clauses 8.1.1 and 8.2);
- routine tests (see Sub-clauses 8.1.2 and 8.3);
- special tests (see Sub-clause 8.1.3).

The tests shall be carried out by the manufacturer at his works, or at any suitable laboratory of his choice.

8.1.1 *Type tests*

They comprise, as appropriate:

- a) verification of temperature-rise limits (see Sub-clause 8.2.2);
- b) verification of dielectric properties (see Sub-clause 8.2.3);
- c) verification of rated making and breaking capacities (see Sub-clause 8.2.4);
- d) verification of ability to carry the rated short-time withstand current (see Sub-clause 8.2.5);
- e) verification of the rated conditional short-circuit current (see Sub-clause 8.2.6);
- f) verification of the rated fused short-circuit current (see Sub-clause 8.2.7);
- g) verification of the rated short-circuit making capacity (see Sub-clause 8.2.8);
- h) verification of the mechanical endurance (see Sub-clause 8.2.9);
- i) verification of the electrical endurance (see Sub-clause 8.2.10).

The type tests, to which a given apparatus shall be submitted according to its kind, are listed in Table VI.

TABLE VI

*List of type tests (referred to by their clause number)
to which a given apparatus shall be submitted*

Nature of test	Switches	Disconnectors	Switch-disconnectors	Fuse-combination units
Temperature-rise limits	8.2.2	8.2.2	8.2.2	8.2.2
Dielectric properties	8.2.3	8.2.3	8.2.3	8.2.3
Rated making and breaking capacities	8.2.4	8.2.4 ¹⁾	8.2.4	8.2.4 ²⁾
Rated short-time withstand current	8.2.5	8.2.5	8.2.5	—
Rated conditional short-circuit current	8.2.6 ³⁾	8.2.6 ³⁾	8.2.6 ³⁾	—
Rated fused short-circuit current	8.2.7 ⁴⁾	8.2.7 ^{4) 5)}	8.2.7 ⁴⁾	8.2.7 ⁵⁾
Rated short-circuit making capacity	8.2.8	—	8.2.8	—
Mechanical endurance	8.2.9	8.2.9	8.2.9	8.2.9
Electrical endurance	8.2.10	—	8.2.10	8.2.10 ⁶⁾

¹⁾ The making and breaking capacity test is only required if a making and/or breaking capacity is stated by the manufacturer (see ²⁾ of Table II).

²⁾ The making and breaking capacity test is only required for fuse-disconnectors and disconnector-fuses if a making and/or breaking capacity is stated by the manufacturer (see ²⁾ of Table II).

³⁾ The rated conditional short-circuit current test is only required if this rated quantity is stated by the manufacturer.

⁴⁾ The rated fused short-circuit current test is only required if this rated quantity is stated by the manufacturer.

⁵⁾ The second test of Sub-clause 8.2.7.2 (making test) is not applicable to disconnectors or fuse-combination units including disconnectors.

⁶⁾ The electrical endurance test is not required for fuse-disconnectors and disconnector-fuses.

8.1.2 Routine tests

They comprise:

- a) operation tests (see Sub-clause 8.3.2);
- b) dielectric tests (see Sub-clause 8.3.3).

8.1.3 Special tests

These are tests subject to agreement between manufacturer and user.

They may include the additional verification of electrical endurance, according to Appendix C (see Sub-clause C-8.4.1).

8.2 Type tests

8.2.1 General

The equipment to be tested shall agree in all its essential details with the drawings of the type which it represents.

Unless otherwise specified or stated by the manufacturer, each type test shall be carried out on an equipment in a clean and new condition.

Unless otherwise stated, the tests shall be made with the same kind of current and, in the case of a.c., at the same rated frequency and with the same number of phases as in the intended service.

For tests, the equipment shall be mounted and installed as indicated by the manufacturer. The details of installation (type and size of enclosure, if any, size of conductors, etc.) shall be part of the test report.

Maintenance or replacements of parts is not permitted, but lubrication and adjustment of the equipment, if prescribed by the manufacturer, is permitted for each type test.

If for convenience of testing, it appears useful to increase the severity of a test (e.g. to adopt a higher frequency of operation in order to reduce the duration of the test), this must not be done without the consent of the manufacturer.

8.2.2 *Verification of temperature-rise limits*

8.2.2.1 *Ambient air temperature*

The ambient air temperature shall be measured during the last quarter of the test period by means of at least two thermometers or thermocouples equally distributed around the equipment at about half its height and at a distance of about 1 m from the equipment. The thermometers or thermocouples shall be protected against air currents, heat radiation and indicating errors due to rapid temperature changes.

8.2.2.2 *Temperature-rise tests of the main circuit*

The equipment shall be mounted approximately as under usual service conditions and shall be protected against undue external heating or cooling.

Equipment having an integral enclosure and equipment only intended for use with a special type of enclosure shall be tested in its enclosure for the rated conventional thermal current test. No openings giving false ventilation shall be allowed.

Details of any enclosure, ventilation arrangements, and sizes of test conductors shall be stated in the test report.

For tests with a.c. single-phase or d.c. currents, the test current shall be not less than the rated conventional thermal current. For tests with multi-phase currents, the current shall be balanced in each phase within $\pm 5\%$, and the average of these currents shall be not less than the rated conventional thermal current.

It is permissible, before beginning the tests, to operate the switch or the mechanical switching device of the fuse-combination unit a few times with or without load or the disconnector without load.

The temperature-rise test of the main circuit is made at the rated conventional thermal current.

Tests on d.c. rated equipment may be made with a.c. supply for convenience of testing, but only with the consent of the manufacturer. Test on a.c. rated equipment shall be made at a frequency of between 45 Hz and 62 Hz where the rated frequency of the equipment is 50 Hz or 60 Hz; for lower or higher rated frequencies, a tolerance of $\pm 20\%$ shall apply.

Fuse-combination units shall be fitted with fuse-links having a rated current equal to the rated thermal current of the combination unit with which the test is made. Details of the fuse-links used for test, i.e. the manufacturer's name and reference, the rated current and the breaking capacity, shall be given in the test report. The type test with the specified fuse-links

shall be deemed to cover the use of any other fuse-link having a power loss, at the rated thermal current of the combination unit, not exceeding the power loss of the fuse-link used for the test.

The test shall be made over a period of time sufficient for the temperature-rise to reach a steady-state value (but not exceeding 8 h). In practice, this condition is reached when the variation does not exceed 1 °C per hour.

Note. — In practice, to shorten the test, the current may be increased during the first part of the test, it being reduced to the specified test current afterwards.

At the end of the test, the temperature-rise of the different parts of the main circuit shall not exceed the values specified in Table V.

Depending on the value of the rated thermal current, the test procedures shall be:

For values of test current up to and including 400 A:

- a) The connections shall be single-core, PVC insulated, copper cables or wires with cross-section areas as given in Table VII.
- b) In the case of multi-pole equipment, tested with a.c., the test may be carried out with single-phase current with all poles connected in series provided magnetic effects can be neglected.
- c) The connections shall be in free air, and spaced at approximately the distance existing between the terminals.
- d) For single-phase or multi-phase tests the minimum length of any temporary connection from equipment terminal to another terminal or to the test supply or to a star point shall be:
 - 1 m for cross-sections up to and including 35 mm² (or AWG 2);
 - 2 m for cross-sections larger than 35 mm² (or AWG 2).

For values of test current higher than 400 A but not exceeding 800 A:

- a) The connections shall be single-core, PVC insulated, copper cables with cross-section areas as given in Table VIII, or the equivalent copper bars given in Table VIII as recommended by the manufacturer.
- b) In the case of multi-pole equipment, tested with a.c., the test may be carried out with single-phase current, with all poles connected in series provided magnetic effects can be neglected.
- c) Cables or copper bars shall be spaced at approximately the distance between terminals. Copper bars shall be finished matt black. Multiple parallel cables per terminal shall be bunched together and arranged with approximately 10 mm air space between each other. Multiple copper bars per terminal shall be spaced at a distance approximately equal to the bar thickness. If the sizes stated for the bars are not suitable for the terminals, or are not available, other bars having approximately the same cross-section and approximately the same or smaller cooling surface may be used. Cables or copper bars shall not be interleaved.
- d) For single-phase or multi-phase tests the minimum length of any temporary connection from the equipment terminal to another terminal or to the test supply shall be 2 m. The minimum length to a star point may be reduced to 1.2 m.

TABLE VII
Standard cross-sections of copper conductors corresponding to the rated conventional thermal current

Range of the rated thermal current (A) ¹⁾	0	7.9	15.9	22	30	39	54	72	93	117	147	180	216	250	287	334
S (mm ²)	1	1.5	2.5	4	6	10	16	25	35	50	70	95	120	150	185	240
Values of the rated thermal current (A) ²⁾	≤6	8 10 12	16 20	25	32	40 50	63	80	100	125	160	200	250	—	315	400

¹⁾ The value of current shall be greater than the value in the first line and less than or equal to the value in the second line.

²⁾ These are standard recommended currents and are given for reference purposes only.

For values of test current higher than 800 A but not exceeding 3 150 A:

- a) The connections shall be copper bars of the sizes stated in Table VIII unless the equipment is designed only for cable connection. In this case, the size and arrangement of the cables to be as specified by the manufacturer.
- b) In the case of multi-pole equipment, tested with a.c., the test may be carried out with single-phase current with all poles connected in series provided magnetic effects can be neglected.
- c) Copper bars shall be spaced at approximately the distance between terminals. Copper bars shall be finished matt black. Multiple copper bars per terminal shall be spaced at a distance approximately equal to the bar thickness. If the sizes stated for the bars are not suitable for the terminals, or are not available, other bars having approximately the same cross-section and approximately the same or smaller cooling surfaces may be used. Copper bars shall not be interleaved.
- d) For single-phase or multi-phase tests the minimum length of any temporary connection from equipment terminal to another terminal or to the test supply shall be 3 m but this can be reduced to 2 m provided that the temperature-rise at the supply end of the connection is not more than 5 °C below the temperature-rise in the middle of the connection length. The minimum length to a star point shall be 2 m.

For values of test current higher than 3 150 A:

Agreement shall be reached between manufacturer and user on all relevant items of the test, such as: type of supply, number of phases and frequency (where applicable), cross-sections of test connections, etc. This information shall form part of the test report.

Note. — In all cases, the use of single-phase a.c. current for testing multi-pole equipment is only permissible if magnetic effects are small enough to be neglected. This requires careful consideration especially for currents above 400 A.

8.2.2.3 *Measurement of the temperature*

The temperature of the different parts shall be measured by means of thermocouples at the nearest accessible position to the hottest spot.

Good heat conductivity between the thermocouple and the surface of the part under test shall be ensured. The thermocouple wire and junction shall be such that they do not influence the temperature rise of the part under test.

8.2.2.4 *Temperature rise of a part*

The temperature rise of a part is the difference between the temperature of this part measured in accordance with Sub-clause 8.2.2.3 and the ambient air temperature measured in accordance with Sub-clause 8.2.2.1.

8.2.2.5 *Corrections*

If the ambient air temperature during the test is between +10 °C and +40 °C, no corrections are necessary to take account of the ambient air temperature during the test and the values of Table V are the limiting values of temperature rise. If the ambient air temperature during the test exceeds +40 °C or is lower than +10 °C, this standard does not apply and the manufacturer and the user shall make a special agreement.

TABLE VIII

Standard test conductors for rated conventional thermal currents higher than 400 A

Value of rated thermal current (A)	Range of rated thermal current (A)	Test connection			
		Cables		Copper bars	
		Quantity	Cross-sections (mm ²)	Quantity	Dimensions (mm)
500	400- 500	2	150 (16)	2	30 × 5 (15)
630	500- 630	2	185 (18)	2	40 × 5 (15)
800	630- 800	2	240 (21)	2	50 × 5 (17)
1 000	800-1 000	—	—	2	60 × 5 (19)
1 250	1 000-1 250	—	—	2	80 × 5 (20)
1 600	1 250-1 600	—	—	2	100 × 5 (23)
2 000	1 600-2 000	—	—	3	100 × 5 (20)
2 500	2 000-2 500	—	—	4	100 × 5 (21)
3 150	2 500-3 150	—	—	3	100 × 10 (23)

Notes 1. — Value of current shall be greater than the first value and less than or equal to the second value.

2. — Bars are assumed to be arranged with their long faces vertical. Arrangements with long faces horizontal may be used if specified by the manufacturer.

3. — Values in brackets are estimated temperature-rises of the test conductors given for reference.

8.2.3 Verification of dielectric properties

8.2.3.1 Condition of the equipment for tests

Dielectric tests shall be made on new equipment mounted approximately as under usual service conditions, including internal wiring, and in a dry condition.

When the base of the equipment is of insulating material, metallic parts shall be placed at all the fixing points in accordance with the conditions of normal installation of the equipment and these parts shall be considered as part of the frame of the equipment. When the equipment is in an insulating enclosure, the latter shall be covered by a metal foil connected to the frame. If the operating handle be metallic, it shall be connected to the frame; if it be of insulating material, it shall be covered by a metal foil connected to the frame.

Equipment not supplied with an enclosure but intended to be used in an enclosure shall be tested in an enclosure stated by the manufacturer to be equivalent to the smallest that is applicable in service.

When the dielectric strength of the equipment is dependent upon the taping of leads or the use of special insulation, such taping or special insulation shall also be used during the tests.

8.2.3.2 Application of the test voltage

The test voltage shall be applied for 1 min as follows:

a) with the main contacts closed:

- 1) between all live parts of all poles connected together and the frame of the equipment;
- 2) between each pole and all the other poles connected to the frame of the equipment.

Note. — Where the base of the equipment is of insulating material, see Sub-clause 8.2.3.1, second paragraph, for the meaning of the word "frame".

On equipment having more than one closed position, the test shall be carried out in each closed position.

b) with the main contacts open:

- 1) between all live parts of all poles connected together and the frame of the equipment;
- 2) between the terminals of one side connected together, and the terminals of the other side connected together.

For the purpose of the above tests, an insulated neutral is to be considered as a pole of the equipment.

On equipment having more than one open position, the test shall be carried out in each open position.

8.2.3.3 Value of the test voltage

The test voltage shall have a practically sinusoidal waveform, and a frequency between 45 Hz and 62 Hz. The source of the test voltage shall be capable of supplying a short-circuit current of at least 0.5 A.

Unless otherwise specified, the value of the test voltage shall be as given in Table IX.

TABLE IX

Rated insulation voltage U_i (V)	Dielectric test voltage (a.c.) (r.m.s.) (V)
$U_i \leq 60$	1 000
$60 < U_i \leq 300$	2 000
$300 < U_i \leq 660$	2 500
$660 < U_i \leq 800$	3 000
$800 < U_i \leq 1 000$	3 500
$1 000 < U_i \leq 1 200^*$	3 500
* For d.c. only	

8.2.4 Verification of rated making and breaking capacities

8.2.4.1 Condition of the apparatus for tests

For enclosed types, the apparatus under test shall be mounted complete in its own enclosure and any apertures normally closed in service shall be closed for the test. For apparatus not supplied with an enclosure but intended to be used in an enclosure, the test shall be carried out in an enclosure stated by the manufacturer to be equivalent to the smallest that is applicable in service. The operating mechanism of the mechanical switching device shall be operated in a manner similar to that stated by the manufacturer as being normal.

Where tests are carried out on fuse-combination units, fuse-links shall be replaced by suitable copper links.

8.2.4.2 Test procedure

The test circuit shall be as described in Sub-clause 8.2.4.3.

The mechanical switching device shall be closed and opened five times with an interval of approximately 30 s between operations.

Note. — For mechanical switching devices of rated thermal current of 400 A or more, the time interval may be increased by agreement between manufacturer and user.

For apparatus of utilization category AC-23, the tests for making capacity and the tests for breaking capacity may be carried out separately on the same apparatus.

During each make-break operating cycle, the mechanical switching device need only stay in the closed position for a period long enough to allow the switching operation to be completed and to enable the current value to be established and the moving parts of the mechanical switching device to come to rest. After each operating cycle, the recovery voltage shall be maintained for at least 0.1 second.

The test voltage and the load shall be applied to the appropriate terminals of the apparatus. For apparatus in which a moving contact remains connected to one of the terminals when the apparatus is in the open position, this test shall be repeated with the supply and load connections interchanged, unless the terminals are specifically and clearly marked for load and supply.

8.2.4.3 Test circuit for the verification of the rated making and breaking capacities

Figures 1, 2 and 3, pages 88, 89 and 90, respectively give the diagrams of the circuits to be used for the tests.

The resistances R_1 and R_2 and the reactances L_1 and L_2 of the test circuit shall be adjusted to satisfy the specified test conditions. The reactors shall be air cored. They shall be always connected in series with the resistors, and their value may be obtained by series coupling of individual reactors. Parallel connecting of reactors is permitted when these reactors have practically the same time-constant.

The test circuit loads L_1 and R_1 shall be connected on the load side of the apparatus.

The prospective short-circuit current at the point of connection to the supply terminals of the apparatus shall be at least ten times the values of the test current, or at least 50 kA, whichever is the lower.

All parts of the apparatus normally earthed in service, including its enclosure, shall be insulated from earth and connected to a point as indicated on Figure 1, 2 or 3. This connection shall include a reliable device D (such as a fuse consisting of a copper wire of 0.1 mm diameter and not less than 50 mm in length) for the detection of the fault current and, if necessary, a resistor limiting the value of the prospective fault current to about 100 A. Any artificial neutral should be substantially inductive, and permit a prospective fault current of at least 100 A.

Apparatus with four poles shall be tested as three-pole apparatus, with the neutral or unused pole connected to the frame. If all poles are identical, one test on three adjacent poles is sufficient; if not, the test shall be repeated on all combinations of three adjacent poles.

The test circuit shall be directly earthed at one point only and the position of this point shall be stated in the test report.

For tests corresponding to utilization categories AC-22 and AC-23, the oscillatory frequency f and the factor γ of the prospective transient recovery voltage shall be adjusted, by means of resistors R_p and/or capacitors C_p in parallel with the load, to the values:

$$f = 2000 \cdot I_c^{0.2} \cdot U_c^{-0.8} \pm 10\%$$

where:

- f is the oscillatory frequency, in kilohertz
- U_c is the rated operational voltage of the apparatus, in volts
- I_c is the breaking current, in amperes.

and $\gamma = 1.1 \pm 0.05$

Note. — The frequency and amplitude of the transient recovery voltage may be adjusted and measured as described in Appendix D of IEC Publication 292-1 for motor starters, but it is not necessary to adjust factor γ or the oscillatory frequency for tests concerning only the making capacity.

For tests corresponding to utilization category AC-21, it is not necessary to adjust the oscillatory frequency of the prospective transient recovery voltage.

The tests corresponding to utilization categories DC-21, DC-22 and DC-23 are under consideration; however, for utilization categories DC-22 and DC-23 the test-circuit load may be replaced by a motor producing the specified current and time-constant values, unless otherwise agreed between manufacturer and user.

Note. — The special case of disconnectors of utilization categories AC-20 and DC-20 having a making and/or breaking capacity is subject to agreement between manufacturer and user.

8.2.4.4 *Test quantities*

The tests shall be performed with values stated by the manufacturer in accordance with Table II of Sub-clause 4.3.11.

The tests shall be taken as valid if the values recorded in the test report differ from the values specified only within the following tolerances:

— Current:	$\pm 5\%$	— Power-factor:	± 0.05
— Voltage:	$\pm 5\%$	— Time-constant:	$\pm 15\%$

For a.c., the tests shall be made at the rated frequency of the apparatus as far as the testing station permits. If this frequency cannot be obtained, the test shall be made at a frequency of between 45 Hz and 62 Hz where the rated frequency of the apparatus is 50 Hz or 60 Hz. For lower or higher rated frequencies, a tolerance of $\pm 20\%$ shall apply.

The power-factor shall be deemed to be the average of the cosines of the angle of phase displacement in all the phases, and the power-factor of any phase shall not vary from the average by more than 25% of that average.

8.2.4.5 *Behaviour of the apparatus during the making and breaking tests*

The apparatus shall perform during the above tests in such a manner as not to endanger an operator or cause damage to adjacent equipment. There shall be no permanent arcing nor flashover between poles or phases, and the fine wire fuse shall remain intact.

The apparatus shall remain mechanically operable and contact welding, such as to prevent an opening operation using normal operating means, is not permitted.

8.2.4.6 *Condition of the apparatus after the making and breaking tests*

After the tests made in accordance with Sub-clause 8.2.4.2, the apparatus shall be capable, without maintenance, of withstanding a voltage equal to twice its rated insulation voltage, applied as stated in Sub-clause 8.2.3.2.

It shall be demonstrated immediately after the test that the mechanical switching device will close and open satisfactorily during a no-load close-open operation.

Note. — A closing operation is considered satisfactory when normal operation of the handle through its full stroke will close the contacts sufficiently for the mechanical switching device to be able to carry its rated operational current.

The contacts shall be in a suitable condition to carry the rated operational current. Where doubt exists, a temperature-rise test shall be carried out on the same apparatus: the measured temperature rises shall not exceed by more than 10 °C the values given in Table V or result in damage to adjacent parts.

8.2.5 *Verification of ability to carry the rated short-time withstand current*

8.2.5.1 *Condition of the apparatus for tests*

The apparatus shall be in the condition specified in Sub-clause 8.2.4.1.

The tests shall be made with the mechanical switching device in the closed position, at any convenient test voltage and starting from the cold state with the apparatus at the ambient air temperature.

8.2.5.2 *Test quantities*

The tests shall be performed at the rated short-time withstand current.

Depending on whether an apparatus is rated a.c. or d.c., it shall be tested as follows. For an apparatus having the same rated current value for a.c. and d.c., the a.c. test shall be taken as valid for the d.c. rating.

a) *For a.c.*

The tests shall be made at the rated frequency of the apparatus with a tolerance of $\pm 25\%$.

The current shall be applied for the time specified in Sub-clause 4.3.7 and its value is the r.m.s. value determined from the oscillogram. This value shall be equal to or higher than the specified value in at least one pole.

The highest peak value of the current during the first cycle of the test shall be not less than n times the rated short-time withstand current, the value of n being as stated in the third column of Table X.

When, however, the characteristics of the testing station are such that the above requirements cannot be obtained, the following alternatives are permitted provided that the integral of the square of the current times the duration, obtained during the test, shall be not less than the product of the square of the rated short-time withstand current and the rated duration of short-circuit:

- 1) If the decrement of the short-circuit current of the testing station be such that the rated short-time withstand current cannot be obtained for the rated time without applying initially an excessively high current, the r.m.s. value of the current may be permitted to fall during the test below the specified value, the duration being increased appropriately, provided that the value of the highest peak current is not less than that specified.
- 2) If, in order to obtain the required peak value, the r.m.s. value of the current has to be increased above the specified current, the duration of the test shall be reduced accordingly.

b) *For d.c.*

The current shall be applied for the specified time and its mean value determined from the oscillogram shall be at least equal to the specified value.

When the characteristics of the testing station are such that the above requirements cannot be obtained for the rated time without applying initially an excessively high current, the value of the current may be permitted to fall during the test below the specified value, the duration being increased appropriately.

Finally, if the testing station is unable to make these tests on d.c., they may, if agreed between manufacturer and user, be made on a.c., provided suitable precautions are taken, for instance, the peak value of current shall not exceed the permissible current.

TABLE X
Relationship between the peak and r.m.s. values of current

R.M.S. value of current (in amperes)	Standard power-factor	Ratio n between the peak and the r.m.s. values of current
$I \leq 10\,000$	0.5	1.7
$10\,000 < I \leq 20\,000$	0.3	2.0
$20\,000 < I \leq 50\,000$	0.25	2.1
$I > 50\,000$	0.2	2.2

8.2.5.3 *Behaviour of the apparatus during the tests*

The apparatus shall behave as specified in Sub-clause 8.2.4.5 where applicable.

8.2.5.4 *Condition of the apparatus after the tests*

After the short-time withstand-current tests, the mechanical parts and the insulation of the apparatus shall be substantially in the same condition as before the tests. Furthermore, the apparatus shall comply with the requirements of Sub-clause 8.2.4.6.

8.2.6 *Verification of the rated conditional short-circuit current*

Under consideration.

8.2.7 *Verification of the rated fused short-circuit current*

8.2.7.1 *Condition of the apparatus for the tests*

The apparatus shall be in the condition specified in Sub-clause 8.2.4.1, but the fuse-links stated by the manufacturer shall be used.

8.2.7.2 *Test procedure*

The test circuit shall be as described in Sub-clause 8.2.7.3.

A first test shall be made on the apparatus, with the mechanical switching device in the closed position, connected in series with fuse-links of the rated maximum current and rated breaking capacity deemed suitable by the manufacturer for use with that mechanical switching device.

A prospective current corresponding to the rated fused short-circuit current of the apparatus shall be applied. The current value is the r.m.s. value determined from the prospective current oscillogram.

When the mechanical switching device is a switch or a switch-disconnector, a second test shall be made with new fuse-links and the test circuit made live with the mechanical switching device open. The mechanical switching device shall be closed onto a prospective current corresponding to the rated fused short-circuit current of the apparatus.

The recovery voltage shall be maintained for at least 0.1 second after interruption of the test current by the fuses.

Details of the fuse-links used for the test, i.e. the manufacturer's name and reference, rated current, rated voltage and breaking capacity shall be given in the test report and, in addition, the Joule-integral and cut-off current obtained during the test. The type test with the specified fuse-links shall be deemed to cover the use of any other fuse-link having a Joule-integral and cut-off current values at the rated fused short-circuit current of the apparatus not exceeding those of the fuse-links used for the test.

8.2.7.3 *Test circuit*

Figures 1, 2 and 3, pages 88, 89 and 90, give the diagrams of the circuits to be used for the tests.

The resistance R_2 and reactance L_2 of the test circuit shall be adjusted to satisfy the specified test conditions. The reactors shall be air-cored. They shall be always connected in series with the resistors, and their value may be obtained by series coupling of individual reactors. Parallel connecting of reactors is permitted when these reactors have practically the same time-constant.

The test circuit loads L_1 and R_1 are not included for the fused short-circuit current tests.

The test circuit shall be directly earthed at one point only and the position of this point shall be stated in the test report.

All parts of the apparatus normally earthed in service, including its enclosure, shall be insulated from earth and connected to a point as indicated on Figure 1, 2 or 3. This connection shall include a reliable device D (such as a fuse consisting of a copper wire of 0.1 mm diameter and not less than 50 mm in length) for the detection of the fault current and, if necessary, a resistor limiting the value of the prospective fault current to about 100 A. Any artificial neutral should be substantially inductive, and permit a prospective fault current of at least 100 A.

Apparatus with four poles shall be tested as three-pole apparatus, with the neutral or unused pole connected to the frame. If all poles are identical, one test on three adjacent poles is sufficient; if not, the test shall be repeated on all combinations of three adjacent poles.

8.2.7.4 *Test quantities*

For a.c., the tests shall be made at the rated frequency of the apparatus as far as the testing station permits. If this frequency cannot be obtained, the tests shall be made at a frequency between 45 Hz and 62 Hz where the rated frequency of the apparatus is 50 Hz or 60 Hz. For lower or higher rated frequencies, a tolerance of $\pm 20\%$ shall apply.

The average value of the power-frequency recovery voltage shall be equal to a value corresponding to not less than 110% and not more than 115% of the rated operational voltage stated by the manufacturer. For a.c., the power-factor of the circuit shall be in accordance with the values in Table X appropriate to the rated fused short-circuit current assigned to the apparatus; for d.c., the time-constant shall be 15 ms, unless otherwise agreed between manufacturer and user. For a.c., the r.m.s. value of the a.c. component of the prospective current in any phase shall be equal to the specified value with a tolerance of $\begin{smallmatrix} +15\% \\ 0\% \end{smallmatrix}$. For d.c., the steady-state value of the prospective current reached shortly after the circuit is closed shall be equal to the specified value with a tolerance of $\begin{smallmatrix} +15\% \\ 0\% \end{smallmatrix}$.

8.2.7.5 *Behaviour of the apparatus during the rated fused short-circuit-current tests*

The apparatus shall behave as specified in Sub-clause 8.2.4.5.

8.2.7.6 *Condition of the apparatus after the rated fused short-circuit-current tests*

After the rated fused short-circuit-current tests, the apparatus shall comply with the requirements of Sub-clause 8.2.5.4.

8.2.8 *Verification of the rated short-circuit making capacity*

8.2.8.1 *Condition of the apparatus for tests*

The apparatus shall be in the condition specified in Sub-clause 8.2.4.1.

8.2.8.2 *Test procedure*

The test circuit shall be as described in Sub-clause 8.2.4.3, but test circuit loads L_1 and R_1 are not included for this test.

The mechanical switching device shall be closed twice with an interval of approximately 3 minutes between these operations onto a prospective peak current not less than the rated short-circuit making capacity of the apparatus. The current shall be maintained for at least 0.1 second.

8.2.8.3 *Test quantities*

The prospective peak current is determined from the calibration oscillogram. In the case of a three-phase a.c. test, it shall be taken as the highest peak value of the three phases.

Note. — For tests on single-pole apparatus, attention is drawn to the fact that the prospective peak making current determined from the calibration oscillogram may differ from the value of the prospective peak making current corresponding to the test, depending on the instant of making.

For a.c., the power-factor shall be within ± 0.05 of the value given in Table X for the r.m.s. value of current corresponding to the prospective peak current. The power-factor of a polyphase circuit is considered as the average of the power-factors of each phase and the power-factor of any phase shall not vary from the average by more than 25% of that average. The average shall be given in the test report.

For d.c., the value of the time-constant shall be 15 ms, unless otherwise agreed between manufacturer and user. For an apparatus having the same rated current value for a.c. and d.c., the a.c. test shall be taken as valid for the d.c. rating.

The frequency shall be the rated frequency of the apparatus with a tolerance of $\pm 25\%$.

The applied voltage shall be equal to 100% of the rated operational voltage of the apparatus, with a tolerance of $\pm 5\%$.

8.2.8.4 *Behaviour of the apparatus during the test*

The apparatus shall behave as specified in Sub-clause 8.2.4.5.

8.2.8.5 *Condition of the apparatus after the test*

After the short-circuit making capacity test, the apparatus shall comply with the requirements of Sub-clause 8.2.5.4.

8.2.9 *Verification of the mechanical endurance*

8.2.9.1 *Condition of the equipment for the tests*

The equipment shall be mounted in a way similar to that expected in service. The operating mechanism shall be operated in a manner similar to normal service operation.

During the test, there shall be no voltage or current in the main circuit.

8.2.9.2 *Test procedure*

The equipment shall be operated for the number of operating cycles specified in Sub-clause 4.3.12. The operating frequency may be any convenient value provided that the moving parts of the mechanical switching device are allowed to come to rest between two successive operations.

8.2.9.3 *Condition of the equipment after the tests*

The condition of the equipment shall be as stated in Sub-clause 8.2.4.6, but some wear on the mechanism parts and the contacts may be allowed provided that the mechanical switching device functions correctly.

Where the design of the equipment permits, maintenance, but not replacement of the contacts, is permissible and under these conditions the equipment shall be able to carry its rated operational current without the temperature-rise limits specified in Table V being exceeded. Where doubt exists, a temperature-rise test at the rated operational current shall be made on the same equipment.

8.2.10 *Verification of the electrical endurance*

8.2.10.1 *Condition of the apparatus for the tests*

The apparatus shall be mounted in a way similar to that expected in service. The operating mechanism shall be operated in a manner similar to normal service operation.

8.2.10.2 *Test procedure*

The apparatus shall be operated for the number of operating cycles stated by the manufacturer in accordance with Sub-clause 4.3.13. If this number is not stated, it shall be 1/20 of the number of the mechanical operating cycles specified in Sub-clause 4.3.12. The operating frequency shall be stated in the test report.

8.2.10.3 *Test circuit*

The circuit shall be as in Sub-clause 8.2.4.3, but it is not necessary to adjust the oscillatory frequency of the transient recovery voltage.

8.2.10.4 *Test quantities*

The tests shall be performed with the values stated by the manufacturer in accordance with Table IV of Sub-clause 4.3.13. The tests shall be taken as valid if the values recorded in the test report differ from the values specified only within the following tolerances:

— Current:	$\pm 5\%$	— Power-factor:	± 0.05
— Voltage:	$\pm 5\%$	— Time-constant:	$\pm 15\%$

8.2.10.5 *Behaviour of the apparatus during the tests*

The apparatus shall behave as specified in Sub-clause 8.2.4.5.

8.2.10.6 *Condition of the apparatus after the tests*

The condition of the apparatus shall be as stated in Sub-clause 8.2.4.6, but some wear on the mechanism parts and contacts may be allowed provided that the mechanical switching device functions correctly.

Where the design of the apparatus permits, maintenance, but not replacement of the contacts, is permissible, and under these conditions the apparatus shall be able to carry its rated operational current without the temperature-rise limits specified in Table V being exceeded. Where doubt exists, a temperature-rise test at the rated operational current shall be made on the same apparatus.

In case of doubt regarding the condition of the silver facing mentioned in the note* of Table V, a temperature-rise test shall be made at the rated thermal current. The temperature rise of the contacts shall not exceed by more than 10 °C the values obtained when a similar apparatus has been tested in accordance with Sub-clause 8.2.2.

8.3 *Routine tests*

8.3.1 *General*

Routine tests shall be carried out under the same, or equivalent, conditions to those specified for type tests in the above clauses.

8.3.2 *Operation test*

A test shall be made to verify the correct mechanical operation of the equipment.

8.3.3 *Dielectric tests*

The tests shall be carried out on dry and clean equipment.

The value of the test voltage shall be in accordance with Sub-clause 8.2.3.3.

The duration of each test may be reduced to 1 s.

The test voltage shall be applied as follows:

- a)* between poles with the main contacts closed (with the main contacts open if there is a shunt circuit between poles);
- b)* between poles and the frame of the equipment with the main contacts closed;
- c)* across the terminals of each pole with the main contacts open.

The use of a metal foil, as specified in Sub-clause 8.2.3.1, is unnecessary.

APPENDIX A

INFORMATION TO BE GIVEN BY THE USER WHEN CONDITIONS FOR OPERATION IN SERVICE DIFFER FROM THE STANDARD

A1. Ambient air temperature

The user shall state to the manufacturer the expected range of ambient air temperature if this temperature can be lower than -5°C or higher than $+40^{\circ}\text{C}$.

A2. Altitude

The user shall state to the manufacturer the altitude of the place of installation if it is more than 2 000 metres (6 600 feet).

A3. Atmospheric conditions

The user shall call to the manufacturer's attention cases where the atmosphere in which the equipment is to be installed may have a relative humidity greater than the values specified in Sub-clause 6.1.3 or contain an abnormal amount of dust, acids, corrosive gases, etc. The same applies if the equipment is to be installed near the sea.

A4. Conditions of installation

The user shall call the manufacturer's attention in case the equipment may be fitted to a moving device, if its support may be capable of assuming a sloping position either permanently or temporarily (equipment fitted aboard ships), or if it may be exposed in service to abnormal shocks or vibrations.

A5. Connections with other apparatus

The user shall inform the manufacturer of the type and dimensions of electrical connections with other apparatus in order to enable him to provide enclosures and terminals meeting the conditions of installation and temperature rise prescribed by this standard, and also to enable him to provide space where necessary to spread out conductors within the enclosure.

A6. Auxiliary contacts

The user shall specify the number and type of auxiliary contacts to be supplied to satisfy requirements such as signalling, interlocking, etc.

APPENDIX B

CLEARANCES AND CREEPAGE DISTANCES

B1. Introduction

It is not possible to lay down a simple set of rules relating to clearances and creepage distances which can be applied to the equipment, as so much depends on variable factors such as atmospheric conditions, the type of insulation employed, the disposition of the creepage paths and the conditions of the system on which the equipment is to be used.

This appendix, therefore, is intended to serve as a guide to the values of minimum clearances and creepage distances to be used. The values given are based on those used in various national specifications, and which are known to give satisfactory service under normal industrial conditions and under system conditions generally found in the majority of the countries in which these specifications are in use.

Further investigation is necessary to obtain a better understanding of the effects of the various factors and thus to determine a more comprehensive set of rules.

B2. General

- B2.1 It is recommended that the surface of the insulating parts should be designed with ridges so arranged as to break the continuity of conducting deposits which may form.
- B2.2 Conductive parts covered only with varnish or enamel, or protected only by oxidation or a similar process, should not be considered as being insulated from the point of view of clearances and creepage distances.
- B2.3 The recommended clearances and creepage distances shall be maintained under the following circumstances:
 - a) On the one hand, without external electrical connections, on the other hand, when conductors insulated or bare, of the type and of any dimensions specified for the equipment, are installed according to the manufacturer's instructions, if any.
 - b) After interchangeable parts have been changed, taking into account maximum permissible manufacturing tolerances.
 - c) Taking into consideration possible deformations due to the effect of temperature, ageing, shocks, vibrations, or due to short-circuit conditions which the equipment is intended to endure.

B3. Minimum values of clearances and creepage distances

The minimum values of clearances and creepage distances to be obtained are given in the following table.

(The table is under consideration.)

B4. Safety requirements concerning isolating distance

(Under consideration.)

APPENDIX C

EQUIPMENT FOR DIRECT SWITCHING OF INDIVIDUAL MOTORS

Switches, switch-disconnectors and fuse-combination units normally intended for direct switching of individual motors shall comply with the additional requirements of this appendix. These requirements are essentially the same as the appropriate clauses in IEC Publication 158-1 (2nd edition) and equipment complying with this appendix may state on the nameplate the appropriate utilization category according to Table C-I.

C-4.3.4 *Rated duty*

Additional rated duties considered as standard are as follows:

C-4.3.4.3 *Intermittent periodic duty or intermittent duty*

Duty in which the main contacts of an equipment remain closed for periods bearing a definite relation to the no-load periods, both periods being too short to allow the equipment to reach thermal equilibrium.

Intermittent duty is characterized by the value of the current, the duration of current flow and by the on-load factor which is the ratio of the in-service period to the entire period, often expressed as a percentage.

Example: An intermittent duty comprising a current flow of 100 A for 4 min in every 10 min may be stated as: "Intermittent duty 100 A, 4 min/10 min" or "Intermittent duty 100 A, 6 operating cycles per hour, 40%".

Standard values of on-load factor are 15%, 25%, 40% and 60%.

C-4.3.4.3.1 *Classes of intermittent duty*

According to the number of operating cycles which they shall be capable of carrying out per hour, equipment is divided into the following classes:

- Class 0.03: up to 3 operating cycles per hour;
- Class 0.1: up to 12 operating cycles per hour;
- Class 0.3: up to 30 operating cycles per hour;
- Class 1: up to 120 operating cycles per hour.

It is recalled that an operating cycle is a complete working cycle comprising one closing operation and one opening operation.

For intermittent duty with a large number of operating cycles per hour, the manufacturer shall indicate, either in terms of the true cycle if this is known, or in terms of conventional cycles designated by him, the values of the rated operational currents which shall be such that:

$$\int_0^T i^2 dt \leq I_{th}^2 \times T$$

where T is the total operating cycle time.

C-4.3.4.4 *Temporary duty*

Duty in which the main contacts of an equipment remain closed for periods of time insufficient to allow the equipment to reach thermal equilibrium, the current-carrying periods being separated by no-load periods of sufficient duration to restore equality of temperature with the cooling medium.

Standard values of temporary duty are 10 min, 30 min, 60 min and 90 min with contacts closed.

C-4.3.5 *Making and breaking capacities*

An equipment is defined by its making capacities and breaking capacities, in accordance with utilization categories as specified in Table C-II (see Sub-clause C-4.3.6).

C-4.3.6 *Utilization category*

The utilization categories as given in Table C-I are considered standard in this standard. Any other type of utilization category shall be based on agreement between manufacturer and user, but information given in the manufacturer's catalogue or tender may take the place of such an agreement.

Each utilization category is characterized by the values of the currents and voltages, expressed as multiples of the rated operational current and of the rated operational voltage, and by the power-factors or time-constants as shown in Table C-II and other test conditions used in the definitions of the rated making and breaking capacities.

For equipment defined by their utilization category, it is therefore unnecessary to specify separately the rated making and breaking capacities as those values depend directly on the utilization category as shown in Table C-II.

The utilization categories of Table C-II correspond in principle to the applications listed in Table C-I.

TABLE C-I
Utilization categories

Category		Typical applications
A.C.	AC-2	Slip-ring motors: Starting, plugging ¹⁾ .
	AC-3	Squirrel-cage motors: Starting, switching off motors during running.
	AC-4	Squirrel-cage motors: Starting, plugging ¹⁾ , inching ²⁾ .
D.C.	DC-2	Shunt-motors: Starting, switching off motors during running.
	DC-3	Shunt-motors: Starting, plugging ¹⁾ , inching ²⁾ .
	DC-4	Series-motors: Starting, switching off motors during running.
	DC-5	Series-motors: Starting, plugging ¹⁾ , inching ²⁾ .
¹⁾ By plugging, is understood stopping or reversing the motor rapidly by reversing motor primary connections while the motor is running. ²⁾ By inching (jogging), is understood energizing a motor once or repeatedly for short periods to obtain small movements of the driven mechanism. <i>Note.</i> — The switching of rotor circuits, capacitors or tungsten filament lamps shall be subject to special agreement between manufacturer and user.		

TABLE C-II

*Verification of rated making and breaking capacities (see Sub-clause C-8.2.4)
Conditions for making and breaking corresponding to the several utilization categories¹⁾*

	Category	Value of the rated operational current	Make			Break		
			I/I_e	U/U_e	$\cos \varphi$	I_c/I_e	U_r/U_e	$\cos \varphi$
A.C.	AC-2	(All values)	4	1.1	0.65	4	1.1	0.65
	AC-3	$I_e \leq 17 \text{ A}$	10	1.1	0.65	8	1.1	0.65
		$17 \text{ A} < I_e \leq 100 \text{ A}$	10	1.1	0.35	8	1.1	0.35
		$I_e > 100 \text{ A}$	8 ²⁾	1.1	0.35	6 ³⁾	1.1	0.35
	AC-4	$I_e \leq 17 \text{ A}$	12	1.1	0.65	10	1.1	0.65
		$17 \text{ A} < I_e \leq 100 \text{ A}$	12	1.1	0.35	10	1.1	0.35
		$I_e > 100 \text{ A}$	10 ⁴⁾	1.1	0.35	8 ²⁾	1.1	0.35
D.C.			I/I_e	U/U_e	L/R (ms)	I_c/I_e	U_r/U_e	L/R (ms)
	DC-2 } (shunt-	(All values)	4	1.1	2.5	4	1.1	2.5
	DC-3 } motors)	(All values)	4	1.1	2.5	4	1.1	2.5
	DC-4 } (series-	(All values)	4	1.1	15	4	1.1	15
	DC-5 } motors)	(All values)	4	1.1	15	4	1.1	15

I_e Rated operational current (see Sub-clause 4.3.2.3);
 U_e Rated operational voltage (see Sub-clause 4.3.1.1);
 I Current made;
 U Voltage before make;
 U_r Recovery voltage;
 I_c Current broken.

¹⁾ For a.c., the conditions for making are expressed in r.m.s. values, but it is understood that the peak value of asymmetrical current, corresponding to the power-factor of the circuit, may assume a higher value.

²⁾ With a minimum of 1 000 A for I or I_e .

³⁾ With a minimum of 800 A for I_e .

⁴⁾ With a minimum of 1 200 A for I .

C-4.3.7 Mechanical endurance

With respect to its resistance to mechanical wear, an equipment is characterized by the number of no-load operating cycles (i.e., without current on the main contacts) which can be made before it becomes necessary to service or replace any mechanical parts; however, normal maintenance including replacement of contacts as specified in Sub-clause 8.2.9.3 is permitted.

The preferred numbers of no-load operating cycles, expressed in millions, are:

0.001 — 0.003 — 0.01 — 0.03 — 0.1 — 0.3 and 1.

If no mechanical endurance is stated by the manufacturer, a class of intermittent duty implies a minimum mechanical endurance corresponding to 8 000 h of operation at the highest corresponding frequency of operating cycles.

C-4.3.8 Electrical endurance

With respect to its resistance to electrical wear, an equipment is characterized by the number of on-load operating cycles, corresponding to the service conditions given in Table C-III, which can be made without repair or replacement.

Unless otherwise stated, the number of on-load operating cycles for categories AC-3, DC-2 and DC-4 shall be not less than 1/20 of the number of no-load operating cycles corresponding to the mechanical endurance of the equipment.

It is recommended that the manufacturer state the above-mentioned number of on-load operating cycles.

TABLE C-III

*Verification of the number of on-load operating cycles
Conditions for making and breaking corresponding to the several utilization categories¹⁾*

	Category	Value of the rated operational current	Make			Break		
			I/I_e	U/U_e	$\cos \varphi$	I_c/I_e	U_r/U_e	$\cos \varphi$
A.C.	AC-2	(All values)	2.5	1	0.65	2.5	1	0.65
	AC-3 {	$I_e \leq 17 \text{ A}$	6	1	0.65	1	0.17	0.65
		$I_e < 17 \text{ A}$	6	1	0.35	1	0.17	0.35
	AC-4 {	$I_e \leq 17 \text{ A}$	6	1	0.65	6	1	0.65
		$I_e > 17 \text{ A}$	6	1	0.35	6	1	0.35
D.C.			I/I_e	U/U_e	L/R (ms)	I_c/I_e	U_r/U_e	L/R (ms)
	DC-2	(All values)	2.5	1	2	1	0.10	7.5
	DC-3	(All values)	2.5	1	2	2.5	1	2
	DC-4	(All values)	2.5	1	7.5	1	0.30	10
	DC-5	(All values)	2.5	1	7.5	2.5	1	7.5
I_e Rated operational current (see Sub-clause 4.3.2.3); U_e Rated operational voltage (see Sub-clause 4.3.1.1); I Current made; U Voltage before make; U_r Recovery voltage; I_c Current broken.								
¹⁾ For a.c., the conditions for making are expressed in r.m.s. values, but it is understood that the peak value of asymmetrical current, corresponding to the power-factor of the circuit, may assume a higher value.								

C-8.2.4 Verification of rated making and breaking capacities

(See Sub-clause 8.2.4, except where stated below.)

C-8.2.4.3 Test circuit for the verification of rated making and breaking capacities

The making and breaking capacity tests shall be carried out with the conventional test circuit as specified in Appendix D of IEC Publication 158-1 (second edition).

C-8.2.4.4 Verification of rated making capacity

The making current to be obtained during the test shall be as given in Table C-II for the appropriate utilization category. The number of closing operations to be made is 20.

The time interval between an opening operation and the closing operation immediately following it shall be 5 s to 10 s.

Note. — For large apparatus, the maximum time interval of 10 s specified above may be increased by agreement between manufacturer and user.

C-8.2.4.5 Verification of rated breaking capacity

The breaking current to be obtained during the test shall be as given in Table C-II for the appropriate utilization category. The total number of opening operations to be made is 25.

The duration of each current flow shall not exceed 0.5 s and the time interval between two successive opening operations shall be 5 s to 10 s.

Note. — For large apparatus, the maximum time interval of 10 s specified above may be increased by agreement between manufacturer and user.

C-8.2.4.6 *Verification of reversibility*

In the case of a reversing switching device, the following test shall be made in addition to the making and breaking capacity tests of Sub-clauses C-8.2.4.4 and C-8.2.4.5. A new switching device may be used for the verification of reversibility.

The test circuit shall be in accordance with Sub-clause C-8.2.4.3, and the current to be obtained shall be as given in Table C-III for category AC-4.

The test comprises ten operating sequences, each sequence comprising the two operating cycles described below:

— 1st cycle: Close A — open A/close B — open B — 10 s to 30 s rest;

— 2nd cycle: Close B — open B/close A — open A — 10 s to 30 s rest

(where A and B are the two circuits of the mechanical switching device).

These cycles are repeated alternately.

The use of a symbolic form such as “open A/close B” implies that the change-over operation concerned shall be made as fast as the normal control system will allow.

During the test, the switching device shall be operated in the manner in which it is intended to be used in service, and any mechanical or electrical interlocking devices which are normally provided shall be in use.

During this test and with the specified number of operations, there shall be no permanent arcing, no flashover between poles, no blowing of the fuse element in the earth circuit (see Sub-clause 8.2.4.3 of this publication) and no welding of the contacts.

C-8.2.9 *Verification of mechanical endurance*

(See Sub-clause 8.2.9 of this standard except that Sub-clause 8.2.9.2 is to be replaced by the following Sub-clause C-8.2.9.2.)

C-8.2.9.2 *Test procedure*

The tests are carried out at the frequency of operations corresponding to the class of intermittent duty. However, if the manufacturer considers that the equipment can satisfy the required conditions when using a higher frequency of operations, he may do so in order to reduce the duration of the tests.

The number of operating cycles to be carried out shall be not less than the number of no-load operating cycles specified in Sub-clause C-4.3.7.

After each tenth of the total number of operating cycles given in Sub-clause C-4.3.7 has been carried out, it is permissible before carrying on with the test:

- to clean the whole equipment without dismantling;
- to lubricate parts for which lubrication is prescribed by the manufacturer for normal service;
- to replace the contacts if they are worn, the wear of the contacts not being taken into consideration during these tests of mechanical endurance.

This maintenance work shall not include any replacement of parts, except for the contacts.

C-8.4 *Special tests*

C-8.4.1 *Verification of electrical endurance*

Note. — Although strictly a type test, this is included under special tests (i.e. subject to agreement between manufacturer and user) because of the difficulty and cost of carrying out electrical endurance tests on some types of equipment. However, it is recommended that, for equipment manufactured in large quantities, the manufacturer shall be prepared to give values for electrical endurance when tested as below.

The currents to be made and broken shall be as given in Table C-III (see Sub-clause C-4.3.8). The test circuit shall comprise inductors and resistors so arranged as to give the appropriate values of current, voltage, power-factor and time-constant. For category AC-4, the test circuit shall be arranged in accordance with Appendix D of IEC Publication 158-1 (2nd edition). In all cases, the speed of operation shall be chosen by the manufacturer.

Tests shall be made with the equipment under the conditions of Sub-clause 8.4.1 of IEC Publication 158-1 (2nd edition), except that, after the test, the equipment shall fulfil the operating conditions specified in Sub-clause 8.2.10.6 of this standard.

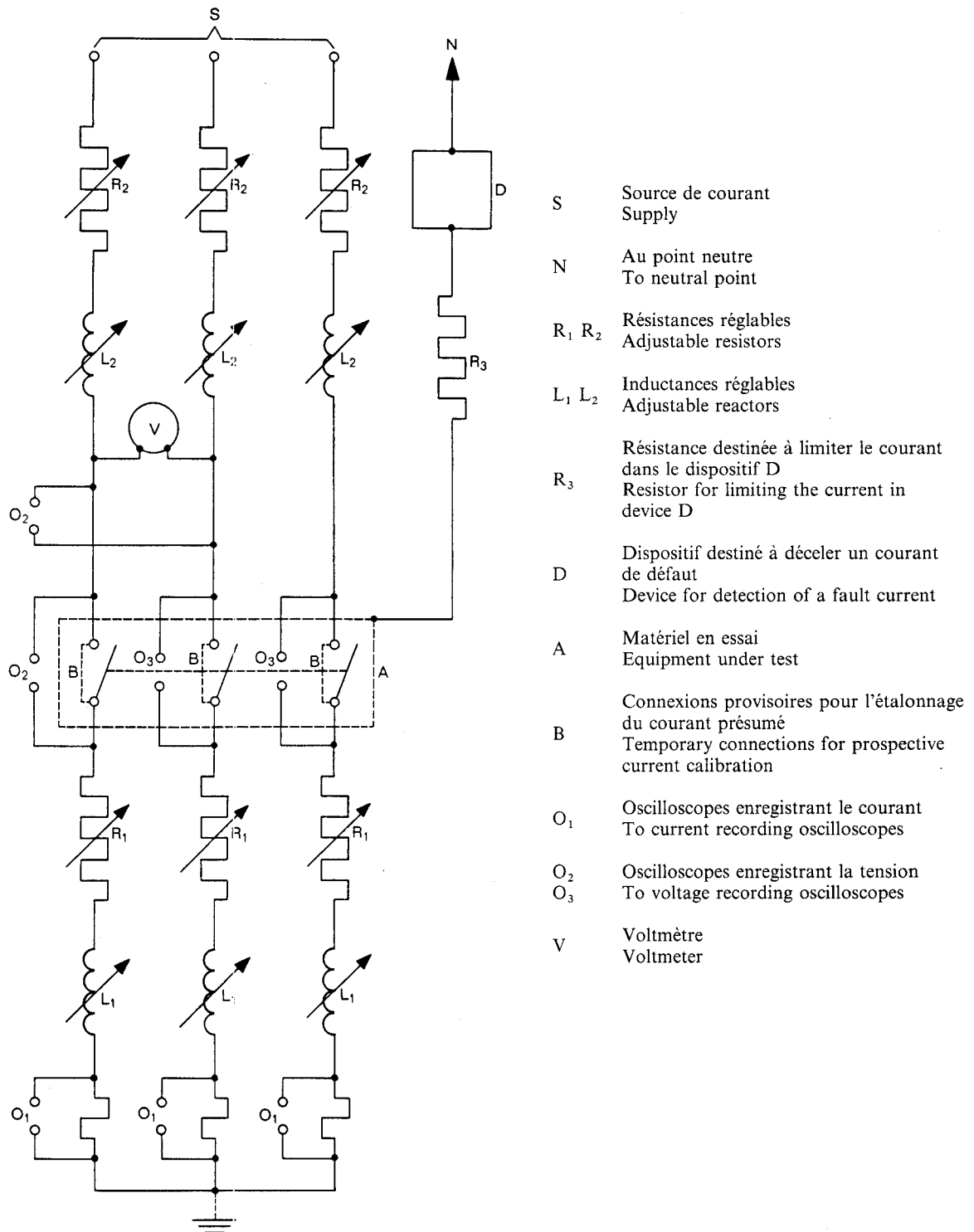
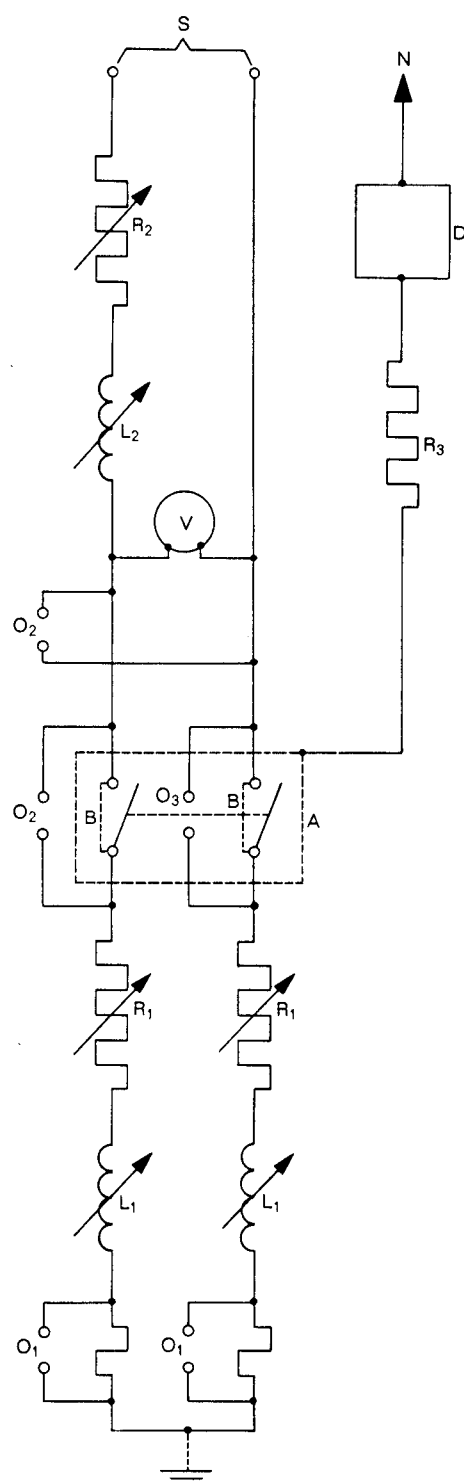
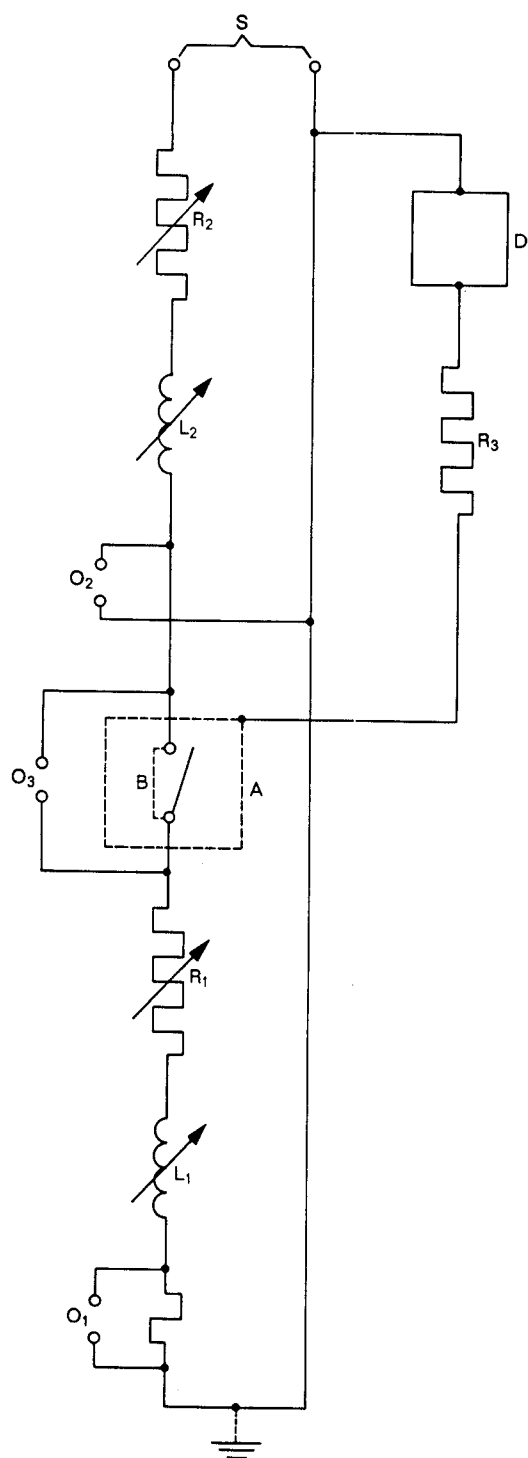


FIG. 1. — Schéma du circuit d'essai pour un matériel tripolaire essayé en courant triphasé.
Diagram of a test circuit for a three-pole equipment on three-phase a.c.



- S Source de courant
Supply
- N Au point neutre ou au point milieu de la source continue
To the neutral point or mid-point of the d.c. supply
- R_1 R_2 Résistances réglables
Adjustable resistors
- L_1 L_2 Inductances réglables
Adjustable reactors
- R_3 Résistance destinée à limiter le courant dans le dispositif D
Resistor for limiting the current in device D
- D Dispositif destiné à détecter un courant de défaut
Device for detection of a fault current
- A Matériel en essai
Equipment under test
- B Connexions provisoires pour l'étalonnage du courant présumé
Temporary connections for prospective current calibration
- O_1 Oscilloscopes enregistrant le courant
To current recording oscilloscopes
- O_2 O_3 Oscilloscopes enregistrant la tension
To voltage recording oscilloscopes
- V Voltmètre
Voltmeter

FIG. 2. — Schéma du circuit d'essai pour un matériel bipolaire essayé en courant monophasé ou en courant continu.
Diagram of a test circuit for a two-pole equipment on single-phase a.c. or on d.c.



- S Source de courant
Supply
- R_1 R_2 Résistances réglables
Adjustable resistors
- L_1 L_2 Inductances réglables
Adjustable reactors
- R_3 Résistance destinée à limiter le courant
dans le dispositif D
Resistor for limiting the current in
device D
- D Dispositif destiné à déceler un courant
de défaut
Device for detection of a fault current
- A Matériel en essai
Equipment under test
- B Connexion provisoire pour l'étalonnage
du courant présumé
Temporary connection for prospective
current calibration
- O_1 Oscilloscope enregistrant le courant
To current recording oscilloscope
- O_2 Oscilloscopes enregistrant la tension
To voltage recording oscilloscopes
- O_3

FIG. 3. — Schéma du circuit d'essai pour un matériel unipolaire essayé en courant monophasé ou en courant continu.

Diagram of a test circuit for single-pole equipment on single-phase a.c. or on d.c.